

Review

Artificial intelligence-powered social robots for promoting physical activity in older adults: A systematic review

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Abstract

Background: The aging global population necessitates innovative strategies to enhance older adults' health and quality of life. Physical activity (PA) is crucial for healthy aging, yet many older adults struggle to exercise regularly. Artificial intelligence (AI)-powered social robots offer an interactive, engaging, and personalized solution to promote PA among this demographic. This systematic review investigated the role of AI-powered social robots in encouraging PA in older adults.

Methods: We conducted a systematic literature search in databases including PubMed, IEEE Xplore, Scopus, Cochrane Library, and Web of Science, focusing on studies published until February 2024. We included peer-reviewed articles reporting empirical findings on designing, implementing, and evaluating AI-enabled social robots to promote PA among older adults. Studies were conducted in nursing homes, rehabilitation centers, community centers, and home environments.

Results: A total of 19 studies were included in the review. Analysis reveals that AI-powered social robots effectively motivate older adults to engage in PAs, leading to increased exercise adherence, higher engagement levels, and extended training durations. Social robots have demonstrated effectiveness across various environments, including nursing homes, rehabilitation centers, community centers, home environments, and elder care facilities. In structured environments like nursing homes and rehabilitation centers, robots help maintain regular exercise routines, improving adherence and recovery outcomes. In community and elder care centers, robots promote PA and social engagement by facilitating group exercises and enhancing participation. In home environments, robots provide personalized support for daily activities, offering reminders and engagement, which fosters long-term activity engagement. User acceptance and satisfaction are high, with participants finding the robots engaging and enjoyable. Additionally, several studies indicate potential health benefits, such as improved medication adherence, better sleep patterns, and enhanced overall well-being. Nevertheless, additional research is imperative to address unresolved issues concerning the technological maintenance costs, design constraints, and adaptability of AI-powered social robots to specific user demographics.

Conclusion: AI-powered social robots play a promising role in promoting PA among older adults, enhancing their health, well-being, and independence. This review provides insights for researchers, designers, and healthcare professionals developing AI-enabled social robotic systems for older adults.

Keywords: Artificial intelligence; Social robots; Older adults; Physical activity; Systematic review

1. Introduction

Regular physical activity (PA) is crucial for older adults as it offers numerous health benefits, such as improved cardiovascular health,¹ enhanced mental well-being,² and better cognitive function.³ The current levels of PA among older adults are

alarmingly low, with many failing to meet the recommended guidelines. Less than 15% of individuals aged 65 and older engage in sufficient PA to maintain health benefits.⁴ This insufficient PA is linked to various adverse health outcomes, including increased risks of cardiovascular disease, type 2 diabetes, and certain types of cancer⁵ and diminished mental health and cognitive decline.⁶ Several challenges hinder older people from engaging in regular PAs, including physical limitations, lack of motivation, and inadequate access to suitable PA programs.⁷ Consequently, there is a pressing societal need

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to develop effective strategies to promote and sustain PA among older adults.

Artificial intelligence (AI) encompasses a range of technologies that enable machines to simulate human intelligence.⁸ Within AI, machine learning (ML) refers to algorithms that learn from data to make predictions or decisions without being explicitly programmed.⁹ Deep learning (DL), a subset of ML, involves neural networks with many layers that can model complex patterns in large datasets.¹⁰ Reinforcement learning (RL) is another subset of ML where agents learn to make decisions by receiving rewards or penalties.¹¹ Social robots, which integrate these AI technologies, are designed to interact with humans socially and intuitively.¹² In the health domain, AI and social robots have been employed in various applications, including disease prediction, health monitoring, and behavior interventions.^{13–15}

AI-powered social robots can provide personalized PA recommendations, deliver real-time feedback, and engage users through motivating interactions.^{16–18} These robots possess the ability to analyze data and adapt interventions according to individual needs, helping older adults overcome barriers to maintaining regular PA.¹⁹ These personalized interventions ensure that the suggested activities are suitable and safe for each individual, considering their physical abilities and health conditions.^{20,21} Furthermore, social robots equipped with AI can also deliver real-time feedback and motivational prompts, helping older adults stay engaged and committed to their PA routines.^{22,23} Real-time feedback allows users to monitor their progress immediately, while motivational prompts assist in maintaining engagement. By providing encouragement and adjusting the difficulty of exercises based on user performance, these robots may boost adherence to PA programs.^{20,24,25} In addition to their technical support, these robots' social interaction capabilities play a crucial role in sustaining older adults' participation. For instance, the robots can provide companionship and encouragement, and through positive reinforcement and adaptive communication, they help create more personalized and human-like interactions, which can further boost users' motivation and adherence to exercise routines.^{26–28} Furthermore, AI-driven data analysis enables the continuous monitoring and assessment of PA interventions, allowing for data-driven decision-making.^{20,29} This approach ensures that PA strategies are constantly refined and optimized, ultimately leading to better health outcomes.³⁰ In summary, the integration of AI and social robots into PA interventions represents a promising advancement that combines the technical precision of AI with the motivational benefits of personalized social interactions. These AI-powered social robots incorporate several "active ingredients" that promote increased PA, such as personalized assistance, real-time feedback, motivation, and interaction, thus demonstrating great potential in promoting sustained PA among older adults.

The existing body of literature on AI-powered social robots for PA interventions is growing,³¹ yet it remains fragmented and varied in its approaches and findings. Several reviews have demonstrated the potential of AI technologies in various fields, but significant gaps persist in understanding their

specific contributions to PA promotion among older adults. For instance, Gabarron et al.³² focused on AI-driven digital solutions, primarily examining recommender systems and conversational agents, but did not address the unique contributions of embodied social robots. Similarly, An et al.³³ emphasized AI methodologies for PA interventions but broadly covered AI technologies without isolating social robots or their interactive capabilities. Oh et al.³⁴ systematically reviewed AI chatbots for lifestyle modifications, including PA, yet their focus did not extend to the physical embodiment or multisensory interactions that distinguish social robots. Moreover, the reviews by Zhang et al.³⁵ and Pu et al.³⁶ emphasize the role of AI social robots in improving psychological well-being and cognitive engagement, and reducing loneliness among older adults. However, neither review specifically addresses the role of social robots in promoting PA, further highlighting a gap in the current literature. The inconsistencies in methodologies, scopes, and focal areas across these studies make it challenging to draw definitive conclusions about the specific role of AI-powered social robots in enhancing PA, particularly in older adult populations.

The aim of this systematic review was to fill this gap by evaluating the effectiveness of AI-powered social robots in promoting PA among older adults, analyzing the different methodologies employed in previous studies, and exploring key factors that contribute to the successful integration of social robots into the daily exercise routines of older adults. This review also focus on the role of social robots in long-term behavior change, explore how different types of social robots may vary in their effectiveness at promoting PA. Additionally, the study investigated user acceptance and technological integration, focusing on the differences and challenges of incorporating social robots into older adults' PA programs in specific environments, such as nursing homes, rehabilitation centers, community centers, and home environments. It aimed to provide actionable insights for researchers, designers, and healthcare professionals to develop more engaging and sustainable AI-powered social robotic systems that enhance the health and well-being of older adults.

2. Methods

This scoping review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guidelines as outlined by Tricco et al.³⁷ and was registered with the International Prospective Register of Systematic Reviews (PROSPERO), adhering to established standards for systematic reviews in healthcare research (Registration No. CRD42024563725).

2.1. Study selection criteria

Studies that met all the following criteria were included in the review: (a) Study design: experimental studies, including randomized controlled trials (RCTs), pre–post interventions; (b) AI methods: AI refers to the technologies that enable machines to simulate human cognitive functions such as learning, reasoning, and problem-solving through the use of

ML, DL, or RL algorithms; (c) AI-powered social robots: use of ML, DL, or RL algorithms, ability to dynamically analyze user data and adjust their behavior in real-time, and application in personalized interventions or decision-making processes support; (d) Study subjects: older adults; (e) outcome variables: the primary outcome variables must include PA measures; (f) Article type: original, empirical, peer-reviewed journal publications; (g) Time window of search: from the inception of electronic bibliographic databases to March 1, 2024; and (h) Language: written in English.

Studies were excluded from the review if they met the following criteria: (a) Non-experimental studies: observational studies, reviews, editorials, letters, case reports, or study protocols; (b) Non-AI social robots: studies that use social robots that are not powered or enhanced by AI; (c) Non-older populations: studies that do not focus on older adults as the primary study population; (d) Non-PA outcomes: studies that did not include PA as a primary outcome variable; and (e) Non-English articles: articles written in languages other than English.

2.2. Search strategy

A comprehensive keyword search was conducted across 5 electronic bibliographic databases: PubMed, Web of Science, Scopus, Cochrane Library, and IEEE Xplore. The search strategy was designed to ensure thorough coverage of relevant studies by combining terms related to social robots and physical activity. The search algorithm utilized for PubMed is detailed in [Supplementary Material 1](#). Two authors (JY and HZ) independently screened the titles and abstracts of the articles identified through the keyword search according to study selection criteria, with a specific focus on identifying whether AI technologies were central to the intervention. Full texts of potentially relevant articles were obtained and reviewed in detail, and studies were assessed for the presence of AI features, such as the use of ML, DL, or RL algorithms. Interventions that incorporated AI-driven decision-making, personalized responses, or real-time behavioral adjustments were deemed relevant. The inter-rater agreement between the 2 authors was assessed using Cohen's κ ($\kappa = 0.78$). A 3rd author (JS) resolved disagreements to ensure consistency and accuracy in the study selection process.

2.3. Data extraction and synthesis

A standardized data extraction form was used to ensure consistency and comprehensiveness in the data extraction process. Two authors (JY and HZ) independently extracted data from each study, including key methodological and outcome variables such as 1st author, publication year, country, study design, sample size and characteristics, health status, female percentage, age, type of social robots, study location or setting, technical specifications and configuration of robots, intervention details, type of exercise, each exercise session duration, exercise frequency, intervention duration, robot's role in exercise, outcome measures, and critical findings. A 3rd author (JS) was involved to resolve disagreements when necessary, ensuring accuracy and consistency

throughout the data extraction process. This systematic approach allowed for a thorough synthesis of the current research on AI-powered social robots for promoting PA in older adults, ensuring that all relevant aspects of the studies were captured and compared.

2.4. Study quality assessment

We used the National Institutes of Health's quality assessment tool for before–after (Pre–Post) studies with no control group to assess the quality of each included study.³⁸ For each criterion met, the study receives 2 points (“Yes”), 1 point for partially meeting the criterion (“Partial”), and no points for not meeting the criterion (“No”). The cumulative score of all criteria generates the overall quality score of the study, ranging from 0 to 24. While this rigorous assessment was instrumental in determining the robustness of the presented evidence, it did not influence the inclusion of studies in our review. Any discrepancies in quality assessments conducted by 2 co-authors (JY and HZ) were resolved through consultation with a third co-author (JS), ensuring integrity and consistency in the appraisal process.

3. Results

3.1. Identification of studies

[Fig. 1](#) presents the PRISMA flow diagram. An initial keyword search identified 6979 articles, and after removing duplicates, 6280 unique articles remained for screening based on their titles and abstracts. Of these, 6234 articles were

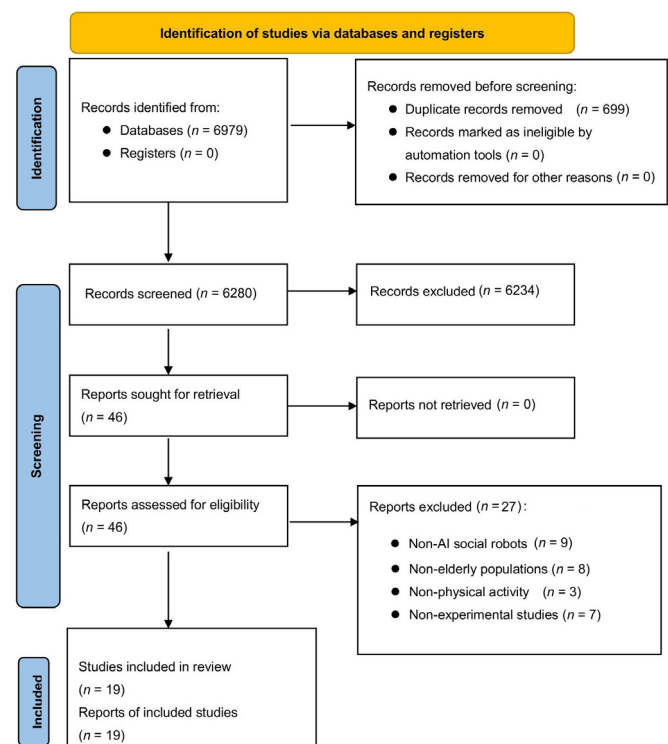


Fig. 1. Preferred reporting items for systematic reviews and meta-analyses (PRISMA) flow diagram.

excluded for not meeting the inclusion criteria. Specifically, studies were excluded because they did not focus on older adults; did not involve AI-powered interventions or social robots; were not related to PA promotion; were reviews, editorials, or commentaries rather than original research; or were not written in English. The study selection criteria were then applied to the remaining 46 articles, excluding 27 studies for various reasons (see Fig. 1 for detailed reasons). Nineteen articles were included in the final review.^{28,39–56}

3.2. Study characteristics

Supplementary Material 2 summarizes the essential characteristics of the 19 studies included in this review on AI-powered social robots for promoting PA among older adults. These studies were published between 2010 and 2023, including 1 study each in 2010,³⁹ 2012,²⁸ 2013,⁴⁰ 2017,⁴³ 2018,⁴⁴ 2019,⁴⁵ and 2022;⁵³ 2 studies^{41,42} in 2016; 3 studies^{50–52} each in 2021 and 2023;^{54–56} and 4 studies^{46–49} in 2020. Fourteen studies^{28,39–43,46,48,49,51,53–56} utilized a Pre–Post design, measuring the outcomes of interest both before and after the intervention within the same group of participants. Five studies^{44,45,47,50,52} employed an RCT design, where participants were randomly assigned to either the intervention group or the control group. Geographically, the studies were conducted in various countries, including the USA ($n = 4$),^{28,39,48,55} Finland ($n = 2$),^{40,56} Japan ($n = 2$),^{41,53} Singapore ($n = 1$),⁴² Germany ($n = 1$),⁴³ New Zealand ($n = 1$),⁴⁴ Sweden ($n = 2$),^{45,47} Italy ($n = 1$),⁴⁶ Switzerland ($n = 1$),⁴⁹ Israel ($n = 1$),⁵⁰ Spain ($n = 1$),⁵¹ Australia ($n = 1$),⁵² and Canada ($n = 1$).⁵⁴ The sample sizes in these studies varied widely, from 4 to 77 participants. Specifically, 6 studies had a sample size between 4 and 11,^{39,41,46,47,49,55} 8 studies had a sample size between 12 and 39,^{28,40,43,45,48,50,54,56} 4 studies had a sample size between 40 and 60,^{42,44,51,52} and 1 study had a sample size above 60.⁵³ The mean sample size was approximately 30 participants, indicating a moderate study size suitable for initial exploratory research in this emerging field. Participants in these studies were predominantly older adults, some with specific health conditions such as chronic obstructive pulmonary disease, dementia, or other mobility and cognitive impairments. The studies were conducted in various settings, including senior living facilities ($n = 2$),^{29,39} nursing homes ($n = 4$),^{40,41,52,56} community centers ($n = 2$),^{42,55} universities ($n = 2$),^{45,47} rehabilitation clinics ($n = 1$),⁴³ participants' homes ($n = 2$),^{44,49} elder care facilities ($n = 3$),^{51,53,54} and laboratory settings ($n = 3$).^{46,48,50} The types of social robots used in these studies varied, including humanoid robots such as NAO,^{40,42,49,50,55,56} Pepper,^{41,45,46,51,54} and Baxter;^{47,48} mobile robotic companions like ROREAS;⁴³ and zoomorphic robots like PARO.⁵² These robots provide assistance, motivation, and interaction to promote PA among older adults.

3.3. Intervention designs and outcome measures

Supplementary Material 3 summarizes the intervention designs and outcome measures of the 19 studies included in the review. These studies utilized various social robots with

distinct technical specifications and configurations to promote PA among older adults. The technical configurations of the robots varied significantly across the studies. Humanoid robots such as NAO, Pepper, and Baxter were most used, and they appeared in 13 studies.^{40–42,45–51,54–56} These robots had advanced features like cameras, microphones, sensors, and audio feedback capabilities. Some studies employed mobile robotic companions like ROREAS and zoomorphic robots such as PARO and PALRO, each designed with specific functionalities like autonomous navigation, emotional and social engagement, tactile sensing, and language-based interactions.^{43,52,53}

Intervention details were diverse, with robots facilitating various types of exercises. The exercises included strength and flexibility training (e.g., upper-limb exercises, shoulder presses), aerobic exercises (e.g., chair aerobics, walking, and gait training), and cognitive-interactive activities (e.g., memory games, interactive touch-based games). The robots' roles encompassed providing instructions, demonstrating exercises, delivering feedback and motivation, real-time monitoring, and social support. Some robots also engaged in co-design workshops for PA, integrating user preferences and limitations into the exercise regimens.

The duration of exercise sessions varied significantly, ranging from short sessions of approximately 10–20 min to longer sessions lasting up to 90 min. Specifically, 6 studies had exercises lasting around 10–20 min per session,^{28,39–42,47} 9 studies had sessions lasting 30–60 min,^{43,45,46,49,52–56} and 1 studies included longer sessions extending up to 90 min.⁴⁸ The frequency of these sessions varied; some interventions involved a single session while others involved multiple sessions per week, ranging from twice a week to daily sessions. The intervention duration across the studies varied as well, from short-term programs of 1–2 weeks in 11 studies,^{28,39,42,43,45–48,51,53,56} medium-term interventions lasting 4–8 weeks in 3 studies,^{52,54,55} and 1 long-term intervention lasting 12–16 weeks.⁴⁴

Overall, the social robots played several roles in these interventions. As exercise instructors, they provided real-time feedback, praise, and motivation, demonstrating exercises and guiding participants through sequences. As social companions, they offered support and engagement, helping participants stay motivated. Some robots also functioned as health monitors, measuring health parameters and reminding participants to take medications. Lastly, robots facilitated various activities, ensuring safety and adapting their behaviors based on user performance and engagement.

3.4. Main findings

Supplementary Material 4 summarizes the impact of AI-powered social robots on promoting PA among older adults.

3.4.1. Effect on PA

The review demonstrates that AI-powered social robots have successfully motivated older adults to engage in PA across various settings, with active ingredients playing a

crucial role. For instance, 2 studies showed that socially assistive robots increased older participants' exercise adherence and enjoyment, especially when participants could choose their exercise games.^{28,39} The active ingredients included personalized feedback, adaptive difficulty levels, and motivational prompts, which helped enhance engagement and progress tracking.^{28,39} Piezzo and Suzuki⁴¹ found that robots serving as walking trainers effectively encouraged older users to walk more. In this case, active ingredients such as step-count monitoring and voice-based encouragement played a key role in maintaining user motivation.⁴¹ Shen and Wu⁴² highlighted that robots were more effective than human instructors in promoting correct joint flexion during exercises. The active ingredients here were real-time visual and auditory guidance, which ensured accurate movement execution.⁴² Meyer and Fricke⁴³ illustrated how robot-assisted training extended the duration and frequency of exercise sessions in stroke rehabilitation patients. In their study, the active ingredients included exercise reminders and physical progress tracking, which motivated patients to stay engaged longer.⁴³ Shao et al.⁵⁴ demonstrated sustained engagement and positive PA outcomes over long-term exercise sessions with robots. Key active ingredients in this case were varied exercise routines and real-time adjustments to the user's performance, which helped maintain long-term interest.⁵⁴ Despite the general positive impacts, the effectiveness of social robots in promoting PA can vary based on settings, user characteristics and design. For example, in structured environments like nursing homes and rehabilitation centers, robots help maintain regular exercise routines, improving adherence and recovery outcomes.^{40,43,56} These settings typically provide more consistent supervision, defined schedules, and access to professional healthcare staff, which enable robots to function effectively as part of a structured regimen. In community and elder care centers, robots promote PA and social engagement by facilitating group exercises and enhancing participation.^{53,54} The group dynamic in these environments fosters social interaction and collective motivation, making robots an effective tool for encouraging engagement through shared activities. In home environments, robots provide personalized support for daily activities, offering reminders and engagement, which fosters long-term activity engagement.^{44,49} The significance of robots lies in their potential to provide personalized support tailored to individual needs and preferences, partially compensating for the lack of direct social support and promoting user autonomy. However, challenges such as physical space limitations, technical issues, and varying levels of user familiarity with technology highlight the need for robots to feature adaptable designs, seamless operation, and highly intuitive, user-friendly interfaces to ensure effective deployment in home settings. Contrasting with other studies, Pu et al.⁵² found no significant effects of social robots on older adults' motor activity. This variability highlights the importance of understanding the contextual factors and specific active ingredients that contribute to PA outcomes.

3.4.2. User acceptance and satisfaction

User acceptance and satisfaction with robot-assisted training are high, with participants generally appreciating the interaction, finding the robots engaging and motivating, and expressing a willingness to use them in the future. Studies revealed that participants viewed robots positively, particularly their relational strategies like praise and empathy.^{28,39,42} High acceptance was also noted in settings like nursing homes⁴⁰ and rehabilitation centers,⁴³ where robots were perceived as helpful and enjoyable. Feedback from studies by Akalin et al.⁴⁵ and Avioz-Sarig et al.⁵⁰ indicated that continuous and combined audio-visual feedback enhanced user engagement and satisfaction, highlighting the importance of personalized and adaptive robot interactions. However, Ruf et al.⁴⁹ reported technical issues that reduced overall satisfaction, and some participants in other studies expressed concerns about the long-term effectiveness and potential habituation effects of robot-assisted training. Cooney et al.⁴⁷ also found that participants appreciated continuous feedback, which was more engaging and fun. However, it is worth noting that the effectiveness and appeal of such feedback may vary depending on the robot's design. For example, humanoid robots, such as Pepper and NAO, with their anthropomorphic features and multimodal interaction capabilities, tend to foster stronger social connections and enhance user engagement, particularly when delivering PA guidance.^{54,55} In contrast, zoomorphic robots, like PARO, leverage their soft, tactile interfaces to facilitate emotional and social engagement, especially in interactions with older adults.⁵² These differing design architectures suggest that robots may offer distinct advantages depending on the context of their deployment.

3.4.3. Effect on health-related outcomes

While the primary focus of the studies was on PA and user acceptance, several indicated potential health benefits, although direct measurements of health-related outcomes were less frequent. Broadbent et al.⁴⁴ reported improved medication adherence and exercise frequency with home care robots, while Pu et al.⁵² observed significant improvements in sleep patterns for residents with dementia and chronic pain. Inoue et al.⁵³ found that robot-assisted leisure programs significantly improved well-being scores, suggesting positive physical and cognitive health impacts. These findings underscore the potential of social robots to promote PA and enhance the overall well-being of older adults. Nonetheless, some study, like Wilson et al.,⁵⁵ found mixed results in health outcomes, indicating that while trends were positive, further research is needed to understand the health impacts of robot-assisted interventions fully. Furthermore, study like Bogliolo et al.⁴⁶ demonstrated the feasibility of using robots for group exercise sessions to maintain participant engagement and prevent or rehabilitate conditions like sarcopenia.

3.5. Study quality assessment

Supplementary Material 5 reports criterion-specific and global ratings from the study quality assessment. The included

studies on average scored 16.3 out of 24 (range: 14–18). All studies included in the review clearly stated the research question/objective and specified and defined the study population. Interventions were consistently detailed across the studies, reflecting a high level of execution fidelity. A majority of the studies ($n=18$)^{28,39,41–56} reliably assessed well-defined, valid, and consistently applied outcome measures. Additionally, excellent retention rates were observed, with 19 studies reporting less than 20% follow-up loss, thus ensuring data integrity.^{28,39–56} However, only 1 study satisfied the criteria for an adequately large sample size,⁵³ and merely half of the studies ($n=8$) conducted comprehensive statistical analyses to rigorously assess the changes in outcome measures.^{28,42,44,45,47,48,50,54} A notable methodological weakness across all studies was the absence of blinding for outcome assessors, potentially introducing bias. Additionally, only 5 studies employed an RCT design, which is a significant limitation.^{44,45,47,50,52} RCTs are considered the gold standard for assessing intervention efficacy, as they provide more robust evidence by controlling for potential confounding factors and reducing selection bias. Future research should prioritize the implementation of RCT designs with larger sample sizes to enhance the validity of the findings. Moreover, none of the studies employed an interrupted time-series design or involved multiple assessments of outcome measures before and after interventions, which could further strengthen the results.

4. Discussion

4.1. AI-powered social robots in promoting PA among older adults

PA promotion among older adults is crucial for maintaining their health, independence, and quality of life, yet many face physical limitations, lack of motivation, and social isolation.^{57,58} AI-powered social robots have emerged as a promising solution to these challenges by providing personalized exercise guidance, encouragement, and companionship.^{20,59} This systematic review reveals that these robots effectively motivate older adults to engage in various physical activities, leading to increased exercise adherence, higher engagement levels, and extended training durations. User acceptance and satisfaction are high, with participants finding the robots engaging and enjoyable. Additionally, several studies indicate potential health benefits, such as improved medication adherence, better sleep patterns, and enhanced overall well-being, all of which highlight the significant potential of AI-powered social robots for improving the quality of life for older adults.

The review demonstrates that AI-powered social robots have successfully motivated older adults to engage in PA. Studies have shown that socially assistive robots can increase exercise adherence and enjoyment, effectively encourage walking, and promote correct joint flexion during exercises.⁴² Additionally, robot-assisted training has been found to extend both the duration and frequency of exercise sessions, maintaining long-term engagement and positive PA outcomes.^{28,39,41–43,54} These findings suggest that robots can

significantly enhance older adults' PA, thereby promoting a healthier lifestyle. However, the effectiveness of social robots in promoting PA can vary across different subgroups, settings, and designs. For example, older adults residing in senior living facilities primarily engaged in light seated arm exercises, which are suitable for individuals with limited mobility.³⁹ Studies focusing on patients with dementia and chronic pain emphasized light activities aimed at providing emotional support and promoting sleep.^{52,53} Meanwhile, older adults with chronic obstructive pulmonary disease participated in tailored pulmonary rehabilitation exercises,⁴⁴ while healthy older adults engaged in strength and balance training, designed to enhance physical fitness and prevent falls.⁵⁵ In the Pu et al.'s study,⁵² motor activity was assessed using objective measures, such as distance walked, step count, time spent in light PA, and energy expenditure. These were recorded through the SenseWear armband device worn by participants during the intervention. While the study primarily focused on sleep, the secondary outcomes included motor activity to determine whether the PARO robot influenced PA levels. However, due to challenges with compliance in wearing the device, particularly in residents with mobility issues, and the lower baseline mobility of participants, no significant effects on motor activity were observed.⁵²

While AI-powered social robots have demonstrated potential for motivating older adults to engage in PA, it is important to compare their advantages and disadvantages with other emerging technologies. Numerous studies outside this review have highlighted increased engagement and adherence to exercise routines facilitated by technological interventions.^{60–62} For example, mobile health (mHealth) Apps and wearable fitness trackers offer convenience and cost-effectiveness, allowing users to easily monitor their activity levels and receive real-time feedback.^{63,64} These devices are widely accessible and can continuously track PA with minimal user effort.⁶⁵ However, mHealth Apps and wearables often lack the personalized, human-like interaction that social robots provide, which can be crucial in motivating older adults, especially those who experience social isolation or cognitive decline.⁶⁶ On the other hand, virtual reality exercise programs offer immersive and engaging experiences, which can significantly enhance motivation and enjoyment, and increase user engagement and PA levels.^{67,68} Building on these strengths, social robots uniquely combine multisensory interactions—verbal, visual, and tactile—to create a highly engaging and personalized user experience. Verbal prompts can encourage users, visual displays can demonstrate exercises, and tactile feedback can offer emotional reinforcement. Despite these advantages, social robots face significant challenges, including high costs, logistical complexities for large-scale deployment, and ongoing maintenance demands. Furthermore, user engagement may decline over time as the novelty of these robots diminishes. These findings underscore the importance of designing interactive and adaptive features into technological interventions to enhance their effectiveness.

Several factors may contribute to the varying effectiveness of social robots in PA promotion reported across studies included in the review. The novelty of the technology could

play a significant role; that is, participants might be more motivated and engaged with newer, more advanced robotic systems compared to older technologies.⁶⁹ The specific design of the robot, including its ability to provide personalized and adaptive feedback, can also influence its effectiveness.^{66,70} Individual differences among participants, such as their initial PA levels, cognitive function, and openness to technology, can affect outcomes.⁶⁰ Understanding these nuances is crucial for optimizing the design and implementation of social robots to maximize their impact on promoting PA among older adults. In addition to these technological and individual factors, the duration and frequency of interventions also play a critical role. Existing studies show significant heterogeneity in these aspects, with the duration of interventions ranging from less than 15 min to as long as 1 h, and the frequency varying from 1-time activities to daily exercises. This heterogeneity presents challenges for assessing the long-term effects of social robots on older adults' PA. We need more interventions with longer durations and consistent frequencies to determine the long-term effects of social robots in promoting PA among older adults.

4.2. User acceptance, engagement, and design factors

The review reveals generally high acceptance and satisfaction rates among older adults using AI-powered social robots for PA. Several factors contribute to this positive reception. First, social robots' relational strategies, such as praise and empathy, significantly enhance user experience.^{28,39,42} Their ability to engage and motivate users also plays a crucial role. Studies like Meyer and Fricke,⁴³ and Akalin et al.,⁴⁵ demonstrated that participants found robot-guided sessions enjoyable and felt more motivated to exercise. Furthermore, the importance of personalized and adaptive interactions cannot be overstated. Cooney et al.⁴⁷ highlighted that continuous and tailored feedback during exercise sessions significantly improved user satisfaction and engagement, reinforcing the need for customization in robotic interventions.

These findings align with existing research on technology acceptance among older adults, which consistently emphasizes the importance of ease of use and perceived usefulness.⁷¹ For example, studies on the acceptance of assistive technologies and mHealth Apps often highlight that older adults are more likely to adopt and consistently use user-friendly technologies that provide clear benefits.⁷²

Despite the high acceptance rates, several challenges persist, particularly regarding technical issues and long-term effectiveness. For instance, Ruf et al.⁴⁹ noted that technical problems such as long startup times, poor facial recognition, and communication delays can harm user experience. These technical barriers are not unique to social robots; similar issues have been reported in studies of other assistive technologies, indicating a broader challenge in the field.^{47,50,73} To enhance user experience, addressing these technical issues through improved design and functionality is essential. Recommendations include ensuring faster response times, enhancing

recognition capabilities, and simplifying the user interface to make interactions more intuitive and seamless.⁷⁴

While many studies report high acceptance, there are instances of conflicting findings where similar technologies have lower acceptance rates. These discrepancies can often be explained by factors such as user training, the novelty effect, or specific features of the robots adopted in the studies. For example, Vaportzis et al.⁷⁵ found that older adults' perceptions of technology could significantly improve with proper training and support, suggesting that inadequate user preparation might lead to lower acceptance. The novelty effect, where new technology initially excites users but wanes over time, can influence acceptance rates. Finally, the specific features and functionalities of the robots, such as their ability to provide engaging and meaningful interactions, play a crucial role in determining user satisfaction. Understanding and addressing these factors is vital for optimizing the design and implementation of social robots to ensure sustained user acceptance and satisfaction.

4.3. Limitations and future research

One of the primary limitations observed in the included studies is the small sample sizes, which can affect the generalizability of the findings. Many studies involved a limited number of participants, which may not adequately represent the broader population of older adults. Additionally, short intervention durations were typical, making it difficult to assess the long-term effectiveness and sustainability of the interventions. Most studies were conducted over a few weeks to a few months, which may not capture the full extent of potential benefits or issues that could arise over a more extended period. Furthermore, there is a potential for biases in self-reported measures, as participants might overestimate their activity levels or satisfaction due to social desirability or the novelty effect of interacting with a robot.⁷⁶ Another notable limitation is the use of Pre–Post study designs without a control group. While such designs allow for the evaluation of changes over time within the same group, they lack a comparison group to account for external factors or natural changes that may occur during the intervention period. This makes it challenging to determine whether observed changes are truly attributable to the intervention or to other influences, such as the placebo effect, participant expectations, or external environmental factors. These limitations highlight the need for future research to include larger, more diverse sample sizes, extended intervention periods, objective measures of PA and user experience, and control groups to provide more robust evidence of the intervention's effectiveness and reduce potential confounding variables.

In addition to the limitations of the individual studies, the systematic review has several constraints. One significant limitation is the potential for publication bias, where studies with positive outcomes are more likely to be published than those with negative or inconclusive results. This bias can skew the overall findings and provide an overly optimistic view of the effectiveness of AI-powered social robots. Another challenge

is the heterogeneity of study designs and outcomes. The included studies varied widely in terms of their methodologies, types of interventions, and measured outcomes, making it difficult to draw definitive conclusions. The diversity in study designs also complicates synthesizing results and identifying consistent patterns, preventing a meta-analysis.

5. Conclusion

The review demonstrates that social robots show promising potential in motivating older adults to engage in PAs, with evidence of increased exercise adherence, higher engagement levels, and extended training durations. High levels of user acceptance and satisfaction further underscore the potential of these technologies to enhance the exercise experience. While direct health-related outcomes were measured less frequently, improvements in medication adherence, sleep patterns, and overall well-being suggest that social robots can positively contribute holistically to the health of older adults, offering both physical and emotional support. However, these results should be interpreted with caution due to the limitations of the current study designs. To confirm these promising findings, future research should address the identified limitations by employing more robust and controlled study designs, extending intervention durations, and utilizing larger and more diverse samples. Additionally, efforts should be made to refine the technology to ensure it is user-friendly, adaptable, and capable of providing personalized and engaging interactions. By doing so, we can advance the development and implementation of AI-powered social robots, ultimately promoting healthier, more active lifestyles for older adults and improving their overall quality of life.

Authors' contributions

JS conceived the study, designed the methodology, and was responsible for project administration; RA contributed to the conceptualization of the study and participated in the review and editing of the manuscript; HZ performed the data curation and contributed to the manuscript revision; JY curated the data and assisted in reviewing and editing the manuscript; ML participated in the manuscript review and editing process. All authors have read and approved the final version of the manuscript, and agree with the order of presentation of the authors.

Competing interests

The authors declare that they have no competing interests.

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Supplementary materials

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